

Semester: VII						
PARALLEL ARCHITECTURE AND GPU PROGRAMMING (Theory and Practice)						
Course Code	:	CS372IA		CIE	:	100+50 Marks
Credits: L:T:P	:	3:0:1		SEE	:	100+50 Marks
Total Hours	:	45L+30P		SEE Duration	:	03 + 03 Hours

Unit-I		9 Hrs
Fundamentals of computer design: Introduction; Defining computer architecture; Dependability, Measuring, reporting and summarizing Performance attributes; Quantitative Principles of computer design Pipelining: Introduction, pipeline hazards Instruction level parallelism (ILP): ILP basic concepts and challenges, basic compiler techniques for exposing ILP, reducing branch costs with prediction, overcoming data hazards with dynamic scheduling, hardware based speculation.		
Unit – II		9 Hrs
Multiprocessors and Thread level parallelism: Introduction, Symmetric shared memory architectures; Performance of symmetric shared memory multiprocessors, Distributed shared memory and directory-based coherence, Basics of synchronization, Models of memory consistency.		
Unit –III		9 Hrs
Data-Level Parallelism in Vector, SIMD, and GPU Architectures: Introduction, Vector Architecture, SIMD Instruction Set Extensions for Multimedia, Graphics Processing Units, Detecting and Enhancing Loop-Level Parallelism. GPU programming with CUDA : Introduction, CUDA's programming model: threads, blocks, and grids, CUDA's execution model: streaming multiprocessors and warps , CUDA compilation process, Putting together a CUDA project. Memory hierarchy:, Local memory/registers, Shared memory , Constant memory, Texture and surface memory		
Unit –IV		9 Hrs
Introduction to Parallel Programming Principles of Parallel Algorithm design: Preliminaries, Decomposition Techniques, Characteristics of Tasks and Interactions, Mapping Techniques for Load Balancing, Methods for containing Interaction Overheads, Parallel Algorithms Models. Programming Using the Using Message Passing Paradigm: Principles of Message Passing Programming, Building Blocks, MPI, Topologies and Embedding , Overlapping Communication with Computation, Collective Communication and computation operations, Groups and Communicators.		
Unit –V		9 Hrs
Parallel Programming in OpenACC OpenACC Syntax, Compute Constructs, Data environment, Loop level parallelism- Kernels Versus Parallel Loops, Three Levels of Parallelism, Other Loop Constructs, Programming Tools for OpenACC – Common Characteristics of Architectures, Compiling OpenACC Code Applied GPU Programming: Application Case Study using pyTorch, TensorFlow, Python Numba		

Course Outcomes: After completing the course, the students will be able to	
CO1:	Understand core concepts of computer architecture to investigate parallel algorithms for solving computational problems.
CO2:	Analyze the performance of parallel programming models and techniques and develop proficiency in parallel algorithm design using MPI, OpenACC, and CUDA.
CO3:	Design and implement parallel algorithms using appropriate parallel programming models
CO4:	Demonstrate Parallel computing concepts for suitable compute intensive real time applications

Reference Books	
1	John L Hennessy, David A Patterson; “Computer Architecture: A Quantitative Approach”, Elsevier, 6 th Edition; 2017, eBook ISBN: 9780128119068, Paperback ISBN: 9780128119051
2	Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar : Introduction to Parallel Computing, Second Edition Pearson Education, 2013, ISBN 13: 9788131708071
3	Sunita Chandrasekaran, Guido Juckeland, OpenACC for Programmers: Concepts and Strategies, Addison- Wesley; 1st edition (9 May 2018), ISBN-13: 978-0134694283
4	Gerassimos Barlas, “Multicore and GPU Programming An Integrated Approach”, Elsevier, 2nd Edition, 2023, ISBN: 978-0-12-814120-5.
5	https://developer.nvidia.com/how-to-cuda-python

LABORATORY COMPONENT	
PART – A	
Students are supposed to execute the programs on computationally intensive algorithms using OpenMP, MPI, CUDA and OpenACC	
PART – B	
Students are supposed to demonstrate a mini project using any of the parallel programming concepts.	

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). Two tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
4.	LAB: Conduction of laboratory exercises, lab report, observation, and analysis (40 Marks) and lab test (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE 50 MARKS	50
MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)		150

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of TWO Sub-divisions only)		
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
Student is required to perform Computer Simulation/ Develop a prototype or model as the case may be and present the results in the form of a presentation. Further, students have to submit a poster for exhibition and also a report.		
1	Design and development of the project	20
2	Presentation of the working model/simulation results/prototype building	20
3	Viva Voce	10
TOTAL		50